

50.021 – Artificial Intelligence

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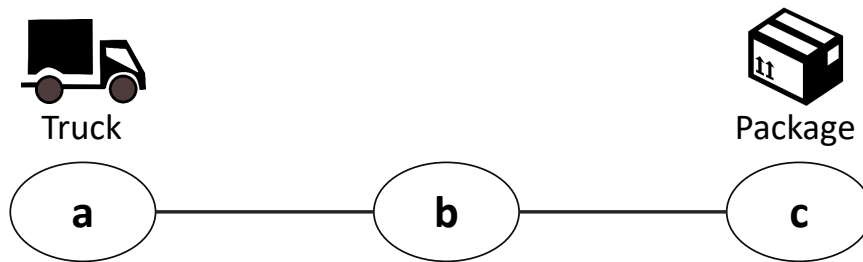
Week 10 Theory Homework - Planning

[The following notes are compiled from various sources such as textbooks, lecture materials, Web resources and are shared for academic purposes only, intended for use by students registered for a specific course. In the interest of brevity, every source is not cited. The compiler of these notes gratefully acknowledges all such sources.]

These answers are provided only as a brief guide. There could be more than one way to answer the questions.

1 Logistic Problem I

Consider the following logistic problem. There are three locations a , b and c , with a truck at a and package at c . The truck is able perform the following actions: (i) $\text{move}(x,y)$: move from location x to y ; (ii) $\text{load}(x)$: load a package at location x ; and (iii) $\text{unload}(x)$: unload the package at location x . The truck can only move between adjacent locations, e.g., a to b , b to c (You can assume that these static facts are already modelled/defined).



Given the start state in the above diagram, your goal is to get the package to location b . Formulate this logistic problem using the STRIPS representation and answer the following:

- List down the propositional variables (facts).
- Specify the operators (actions), including the pre-conditions and post-conditions.
- Specify the initial state.
- List down the goal state/specification.

Q1 Answers:

a.) Facts: truckAt(x), packAt(x), packInTruck

b.)

```
(:action  move(x,y)
      :preconditions  truckAt(x)
      :postconditions  not truckAt(x), truckAt(y)
)
(:action  load(x)
      :preconditions  truckAt(x), packAt(x)
      :postconditions  not packAt(x), packInTruck
)
(:action  unload(x)
      :preconditions  truckAt(x), packInTruck
      :postconditions  packAt(x), not packInTruck
)
```

c.) Initial State: truckAt(a), packAt(c)

d.) Goal State: packAt(b)

2 Logistic Problem II

Based on your STRIPS formulation from Q1 (Logistic Problem I), answer the following:

- a.) What is the optimal solution to this problem?
- b.) Make this a delete-relaxed problem. What are the changes to the original STRIPS formulation you made?
- c.) Based on this delete-related problem, list down all the facts F_x and actions A_x at levels $x = \{0, 1, \dots, M\}$.

Q2: Answers:

a.) move(a,b), move(b,c), load(c), move(c,b), unload(b)

b.) any answer that removes all delete postconditions from the actions

c.)

F0: truckAt(a), packAt(c)

A0: move(a,b)

F1: truckAt(a), packAt(c), truckAt(b)

A1: move(b,c) [also ok if move(b,a) is listed]

F2: truckAt(a), packAt(c), truckAt(b), truckAt(c)

A2: load(c) [also ok if move(c,b) is listed]

F3: truckAt(a), packAt(c), truckAt(b), truckAt(c), packInTruck

A3: unload(a), unload(b), unload(c)

F4: truckAt(a), packAt(c), truckAt(b), truckAt(c), packInTruck, packAt(a), packAt(b)

3 Logistic Problem III

Based on your answer from Q2 (Logistic Problem II), answer the following:

- What is the optimal solution to this delete-relaxed problem? What is this heuristic called?
- What is the value of h_{add} ? Explain why.
- What is the value of h_{max} ? Explain why.

Q3: Answers

a.) `move(a,b), move(b,c), load(c), unload(b)`. This is the h^+ heuristic.

b.) $h_{add} = 4$. Adds cost of all goal facts up. In this case, there is only 1 goal fact, `packAt(b)`, which is reached at F4 (see Q2c).

c.) $h_{max} = 4$. Max cost out of all goal facts. In this case, there is only 1 goal fact, `packAt(b)`, which is reached at F4 (see Q2c).

4 Generic Planning I

Consider a STRIPS problem with propositional variables (facts) m, n, o, p , and the below STRIPS actions with their pre/post-conditions.

Action	Pre	Add	Del
A	m	n,o	$\emptyset$
B	m,o	p	m
C	p	m	p
D	n,o	p	o

Given an initial state $s = \{m\}$ and goal specification $g = \{m, n, o, p\}$, answer the following questions:

- What is the value of h^+ ? Explain why.
- What is the value of h_{add} ? Explain why.
- What is the value of h_{max} ? Explain why.

Q4: Answers:

F0: m

A0: A

F1: m, n, o

A1: B, D

F2: m, n, o, p

a.) $h^+ = 2$. Solution (cost) to delete-relaxed version of problem (see above facts and actions)

b.) $h_{add} = 4$. Adding up cost of all goal facts, i.e., $0+1+1+2$.

c.) $h_{max} = 2$. Max cost of all goal facts, i.e., 2

5 Generic Planning II

Based on the same STRIPS formulation in Q4 (Generic Planning I). Now, based on initial state $s = \{p\}$ and goal specification $g = \{m, n, o, p\}$, answer the following questions:

- a.) What is the value of h_+ (if any)? Explain why.
- b.) What is the value of h_{add} (if any)? Explain why.
- c.) What is the value of h_{max} (if any)? Explain why.

Q5: Answers:

F0: p

A0: C

F1: p, m

A1: A

F2: p, m, n, o

- a.) $h_+ = 2$. Solution (cost) to delete-relaxed version of problem (see above facts and actions)
- b.) $h_{add} = 5$. Adding up cost of all goal facts, i.e., $0+1+2+2$.
- c.) $h_{max} = 2$. Max cost of all goal facts, i.e., 2